LESSON 8.5
FINDING THE MISSING DIMENSION OF A COMPOSITE FIGURE

LESSON INTRODUCTION

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

LEARNING TARGETS

- I can find the value of a missing dimension for a quadrilateral or a composite figure.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

WORKING TOWARD

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

ESSENTIAL QUESTIONS

- What strategies can you use to find the missing dimension on a composite figure?
- How do you use given dimensions to find the missing dimension of a composite figure?
- How can you use known math facts and solving for unknown values in equations to find a missing dimension on a composite figure?

LESSON NARRATIVE

In this lesson, students begin by reviewing math facts that will be used to think about how to find the missing dimension of a rectangle or triangle. Students are guided to use the area of a composite figure to solve for missing dimension lengths on the figure, using strategies to decompose the composite figure and solve for an unknown variable using equations for total area. The 8.5.6 Wrap-Up includes a self-evaluation to explain what students still do not understand and to ask questions about the lesson.

COMPONENT SUMMARY

8.5.1 WARM-UP *(5 MIN.)*

Students create a triangle on a coordinate plane closest to six square units

8.5.2 COLLABORATION *(10 MIN.)*

Collaboration to find missing math facts in equations and missing dimensions in rectangles and triangles

8.5.3 GUIDED INSTRUCTION *(10 MIN.)*

Guided instruction on relationship between the area and shapes within the composite figure

8.5.4 GUIDED PRACTICE *(10 MIN.)*

Guided practice finding missing dimension of a composite figure using the given measurements

8.5.5 YOUR TURN! *(10 MIN.)*

Individual practice finding missing dimension in composite figure using given measurements

8.5.6 WRAP-UP *(~5 MIN.)*

Reflection on lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Dimension
- Composite figure
- Area

MATERIALS

- (Optional) Scratch, tracing, or graph paper
- (Optional) Scissors

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- When more than one side does not have a measurement already labeled, students may struggle with discovering which side they need to solve for.
- Students with gaps in spatial awareness may struggle with decomposing a composite figure.
- Students may incorrectly think that the area of a composite figure is found by adding all the sides or only multiplying all the sides.

THINGS TO CONSIDER

- Encourage students to decompose composite figures in different ways to find the best way to calculate the missing dimension.
- Consider not giving students a four-function calculator for this lesson to build students' fluency on the operations and using rational values.
- Consider giving students a number or two they could use in the 8.5.1 Warm-Up to get started.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 8.5.6 Wrap-Up, students engage in a reflection to think about and then discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts related to solving for missing dimensions on composite figures.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

During 8.5.2 Collaboration, students apply their knowledge of math facts to find missing factors and area of rectangles and triangles to find missing dimensions on composite figures. Using background knowledge and discussing multiple approaches to solving for area or missing dimensions can position students to complete tasks accurately and with confidence.

DIFFERENTIATE INSTRUCTION

In the 8.5.3 Guided Instruction and 8.5.4 Guided Practice portions of this lesson, students need to make sense of visual representations of composite figures to solve for the missing dimension. Consider providing visual and kinesthetic entry points to learning, including:

- Graphing the given figures by hand on graph paper or using online tools like GeoGebra to get a better sense of how the side lengths relate to each other.
- Providing graph paper for students to recreate the drawing to better understand how to decompose the shape.
- Using tracing paper to trace, cut out, and manipulate composite figures; physically cutting paper to decompose the shapes and identify how to use given measurements to find missing dimensions.
- Using graphic organizers or lists to keep track of steps for solving or different equations used to find area of smaller shapes within composite shape.

8.5.1 WARM-UP: CLOSEST TO SIX

Component Overview: *Students create a triangle on a graph whose area is closest to where the digits one through nine are used only one time each as coordinates. Consider using Think-Pair-Share routines on this activity, as students explore the area of a triangle and how it is related to a rectangle.*

TEACHER MOVES

Think-Pair-Share

Consider using Think-Pair-Share routines as students test and determine an area of a triangle closest to six square units. As students share with partners, be sure to prompt students to think about what the possible side lengths are needed to get an area of 6. When students share ideas with a class, write responses on a shared classroom whiteboard so the whole class can see ideas and make connections or identify solutions that may not work.

DIFFERENTIATE INSTRUCTION

As a visual approach for students struggling to get started, consider having students construct a rectangle whose area is equal to 12 square units using graph or tracing paper, then cutting it in half to represent 6 square units. Then have students move the triangle around the graph until they find points that are numerically different and cover approximately the same area.

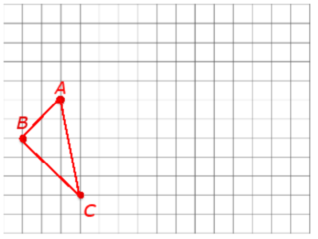
8.5.1 Warm-Up: Closest to Six

1. Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.

A (3 , 6)

B (1 , 5)

C (4 , 2)

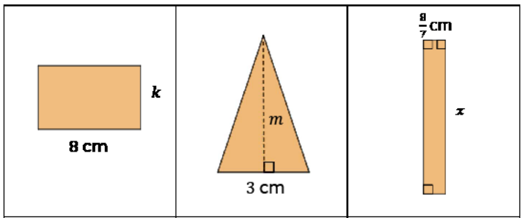


8.5.2 Collaboration: Finding a Missing Value

1. Complete the table by filling in the blanks for the following math facts.

If.....	Then....
$40 \div 8 =$ _5_	$8 \times$ _5_ $= 40$
$27 \div 3 \div \frac{1}{2} =$ _18_	$\frac{1}{2} \times$ _18_ $\times 3 = 27$
$8 \div \frac{8}{7} =$ _7_	_7_ $\times \frac{8}{7} = 8$

2. With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).



Area = 40 cm^2	Area = 27 cm^2	Area = 8 cm^2
$k =$ _5_	$m =$ _18_	$x =$ _7_

8.5.2 COLLABORATION: FINDING A MISSING VALUE

Component Overview: *Students use operations reasoning to find a missing value in a multiplication or division equation, then determine how it relates to an equation using the same numbers in a different order with a different operation. Students then explore how to transfer this knowledge to finding the missing side length of a shape. Consider partnering students to engage in math discussions and check answers as they collaborate.*

ENRICHMENT

For question 1, encourage students to make connections between prior knowledge and the geometry tasks in this section by asking students:

- Which operations are inverses?
- How can inverse operations be used to find a missing number?

DIFFERENTIATE INSTRUCTION

Use visual approaches to see the connection between the formula and diagrams shown by having students write the area formulas for each shape and substitute the given information until they are left with a variable to solve for.

ENRICHMENT

For question 2, prompt student thinking about the relationship between the shapes' areas and missing dimensions by asking:

- What is the relationship between the area of a shape and the given dimension?
- How do you find the area of a triangle and rectangle?
- Where in the real world would you need to find a missing dimension?

8.5.3 GUIDED INSTRUCTION: FINDING A MISSING DIMENSION

Component Overview: *Students explore the relationship between finding the area of shapes that make up the composite figures and how area can be used to find other measurements. Consider having larger versions of the composite shapes displayed for the whole class to see as the class works together to decompose a composite figure using rectangles and triangles, then solve for the missing dimension using the inverse operation.*

DIFFERENTIATE INSTRUCTION

For this section, consider a foldable, Frayer Model, or type of interactive notebook for students to record and summarize their learning on finding the area of composite figures and missing dimensions. Examples, non-examples, and verbal descriptions can provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

ENRICHMENT

Use the questions to scaffold further discussion for questions 1 and 2:

- What do "sum" and "difference" mean?
- How do you find the area of a rectangle?
- How could you write the area of the composite figure as two separate equations? As one combined equation?
- Why is it necessary to know the dimension of each rectangle to find the composite figure area?

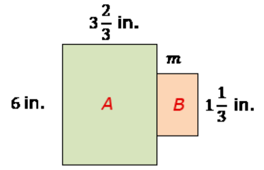
QUESTIONING

Prompt early finishers to create their own composite figure made up of two or more rectangles or triangles, write measurements for each side, and find the area. If time permits, students could then erase one of the dimensions and have a partner try to solve for the missing dimension value given the other measurements and area.

8.5.3 Guided Instruction: Finding a Missing Dimension

To find the area of a composite figure, find the sum of the decomposed parts of the figure.

The figure shown is composed of two rectangles.



1. **Label the rectangle on the left A, and the rectangle on the right B.**

The area of the figure is 26 square inches (in²).

2. **Complete the statements.**

   The area of the composite figure is the ○ sum / ○ difference of the area of Rectangle A and the area of Rectangle B.

   Therefore, the area of the composite figure ○ plus / ○ minus the area of Rectangle A is equal to the area of Rectangle B.

3. **The area of the composite figure is __26__ in².**

4. **The area of Rectangle A is __22__ in².**

5. **Find the area of Rectangle B.**
   $4\ in^2$

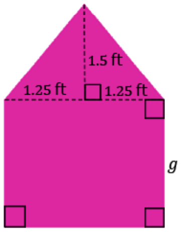
6. **Complete the statement.**

   The missing dimension of Rectangle B can be determined by ○ multiplying / ○ dividing the ○ area / ○ height of Rectangle B by the given dimension.

7. **What is the missing dimension, m, of Rectangle B?**
   $m = 3\ in$

8.5.4 Guided Practice: Finding the Value of the Missing Dimension

1. Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of 6.875 square feet (ft²).

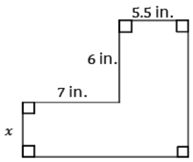


   a. What is the area of the triangular top? Include units.
   $1.875\ ft^2$

   b. What is the area of the rectangular base? Include units.
   $5\ ft^2$

   c. What is the height, g, of the rectangular base? Include units.
   $2\ ft$

2. The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.



$x = 4.08\ in^2$

8.5.4 GUIDED PRACTICE: FINDING THE VALUE OF THE MISSING DIMENSION

Component Overview: *Students work individually to practice finding the missing dimension of a composite figure using given measurements. This activity combines prior knowledge of how to find a missing value in an equation with using an area equation to find the missing dimension.*

ENRICHMENT

As students engage in question 1, provide questions to scaffold student thinking and reasoning about composite figures and finding area using properties of operations.

- How do you find the area of a triangle?
- Which side is the base of the triangle? How can you tell?
- What can you multiply by 1.25 to get the base of the triangle measurement? What's another expression you could use with addition?
- What is the formula for area of a triangle? What happens when you find ½ of 2(1.25)?
- How can you use this information to simplify your work before multiplying?

DIFFERENTIATE INSTRUCTION

Students may incorrectly label sides using given measurements instead of solving for the unknown side lengths. Consider providing string as a measuring tool for students to verify measurements.

TEACHER MOVES

Contemplate, then Calculate

Prompt students to consider their approach to solving before calculating. Infusing this metacognitive strategy will contribute to fluency and mitigate errors.

8.5.5 YOUR TURN!

Component Overview: *Students demonstrate their understanding of how to find a missing dimension using previous knowledge of how to find the area of a composite figure and how to find a missing factor in an equation. Consider having students work individually as a formative assessment on this lesson's learning targets. Note that for both questions, the composite figure will need to be decomposed and the sums set equal to the area of the composite figure to find the missing dimensions.*

DIFFERENTIATE INSTRUCTION

As a kinesthetic entry point, consider giving students a sheet of paper to draw the shapes that compose the composite figure. Have students cut out the shapes to visualize which measurement is missing. Also consider having students graph the given figures by hand or on GeoGebra to get a better sense of how the side lengths relate to each other.

ENRICHMENT

For individual students who may have difficulty getting started, initiate geometric reasoning and problem-solving by asking:

For question 1:

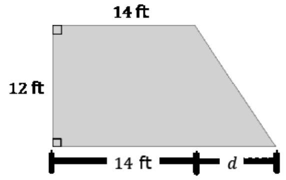
- What two shapes can you decompose the composite figure into that you know how to find the areas of?
- What dimension on the right triangle is also 12 feet tall? How can you tell?

For question 2:

- How did you approach decomposing this figure? What were your reasons for decomposing it that way?
- How can you tell what the height of the triangle is? If you know the height and you are solving for the missing dimension c as the base, what other value do we need to solve for c?

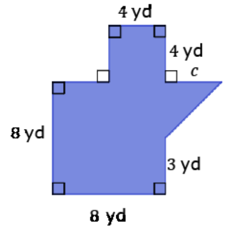
8.5.5 Your Turn!

1. **Julio is building a new patio at his house. He needs 216 square feet (ft^2) of material to cover the area of his design, as shown. Find the missing dimension. Include units.**



$d = 8\ ft$

2. **The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.**



Not drawn to scale

$c = 3\ yd$

8.5.6 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about ...
   Answers will vary.

2. One question I still have about today's lesson is ...
   Answers will vary.

8.5.6 WRAP-UP: REFLECTION

Component Overview: *This Wrap-Up positions students to reflect on the lesson's learning targets and provide an opportunity to ask remaining questions about finding area of composite figures and/or finding missing dimensions. Consider using an auditory, collaborative approach to this reflection by having students share their questions with a partner and seek answers to clarify any misconceptions.*

TEACHER MOVES

Think-Pair-Share

After students have had a few minutes to consider their responses, allow students to turn and talk to discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts for themselves or others.

